A One-Node Counterexample to Strong Arc-Consistency

Inter-witness pair-domains are necessary for $\mathcal{ALCI}_{\mathsf{RCC5}}$ realisability

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Disclaimer. This document was generated by Claude (Anthropic), prompted by Michael Wessel. It corrects a conjecture made in an earlier counterexample paper by the same author [2]. The results have not been independently verified. We invite scrutiny and corrections.

Abstract

In a recent two-node counterexample paper [2] we refuted the naive typed-patchwork conjecture of [1] and proposed a stronger propagation—*strong arc-consistency* (Def. 5.1)—with a companion conjecture (Proposition 5.2) that strong arc-consistency is complete for realisability. We did not prove the companion; we noted the proof would amount to a repackaging of the split-forest soundness theorem. This paper refutes the companion conjecture too. We exhibit a *one-node* TBox and network that is strongly arc-consistent but unrealisable: two pending existentials at the same node force witnesses whose pairwise relations are all type-unsafe. Strong arc-consistency checks each existential against the existing network but does not check pending existentials against *each other*. The minimal fix is inter-witness pair-domains, which is exactly what the split-forest paper's rank-d validity conditions encode [3]. Combined with the two previous papers, this completes a three-step cascade: every simplification of split-forest validity we have tried has a counterexample strictly smaller than the last, and the convergence point is split-forest validity itself.

1 Introduction

This is the third in a cascade of papers attempting to find a simpler propagation complete for $\mathcal{ALCI}_{\mathsf{RCC5}}$ realisability than the split-forest paper's rank-d validity conditions.

- **Patchwork probe** [1] conjectured (Proposition 7.1) that *weak* arc-consistency (composition + type-safety over existing V) is complete.
- **Two-node counterexample** [2] refuted Proposition 7.1 with a two-node, seven-axiom example and proposed *strong* arc-consistency (Def. 5.1): weak plus existential-witness extensibility against every existing $z \in V$. Companion Proposition 5.2 conjectured this stronger form is complete.

- **This paper** refutes Proposition 5.2 with a *one-node* counterexample. Strong arc-consistency checks each pending existential against existing nodes but not against *other pending existentials at the same or different nodes.* The minimal next fix is inter-witness pair-domains.

The cascade converges: each simplification we tried has a strictly smaller counterexample (7 axioms $\to$ 7 axioms, 2 nodes $\to$ 1 node), and each correction adds exactly the check the previous definition missed. The convergence point is the split-forest paper's rank-d validity conditions, which include pair-domains for every pair of (existing or pending) rank-d states.

Structurally: *the split-forest rank-d validity is not cosmetic; it is the minimum viable propagation.*

What this does and does not say. Same caveat as the two-node paper. This is a correction to a specific conjecture. It does not falsify $\mathcal{ALCI}_{\mathsf{RCC5}}$ decidability. The split-forest soundness theorem [3] already uses inter-witness pair-domains (via rank-d witness menus); only our simplified propagation definitions miss them.

2 Strong arc-consistency recalled

Definition 2.1 (Strong arc-consistency; [2], Def. 5.1)**.** A typed network (V, D, τ) is *strongly arc-consistent* if it is weakly arc-consistent (composition + type-safety fixed-point with non-empty domains over V) and, in addition: for every $x \in V$ and every $\exists R.E \in \tau(x)$, there exists $\gamma \in \mathrm{Tp}(C_0, T)$ with $E \in \gamma$ and the forward-propagated universals $\{F : \forall R.F \in \tau(x)\} \subseteq \gamma$, such that for every $z \in V \setminus \{x\}$:

$$\mathrm{comp}(\mathrm{inv}(R), D(x, z)) \cap \mathrm{TypeSafe}(\gamma, \tau(z)) \neq \varnothing.$$

The existential clause iterates the check over every $z \in V \setminus \{x\}$—that is, over every *existing* node other than x. It does not iterate over other *pending* existentials at x itself, nor at any other node. We now show this is a problem.

3 The one-node counterexample

Example 3.1 (Two conflicting existentials at one node)**.** Let T be the $\mathcal{ALCI}_{\mathsf{RCC5}}$ TBox with concept names A, D_1, D_2:

$$\begin{aligned}
\text{(T1)} \quad & A \sqsubseteq \exists \mathsf{PP}.D_1, \\
\text{(T2)} \quad & A \sqsubseteq \exists \mathsf{PP}.D_2, \\
\text{(T3)} \quad & D_1 \sqsubseteq \forall \mathsf{DR}.\neg D_2, \\
\text{(T4)} \quad & D_1 \sqsubseteq \forall \mathsf{PO}.\neg D_2, \\
\text{(T5)} \quad & D_1 \sqsubseteq \forall \mathsf{PP}.\neg D_2, \\
\text{(T6)} \quad & D_1 \sqsubseteq \forall \mathsf{PPI}.\neg D_2, \\
\text{(T7)} \quad & D_1 \sqsubseteq \neg D_2.
\end{aligned}$$

Take $C_0 = A$ and $V = \{x\}$ with $\tau(x)$ a Hintikka type for (C_0, T) containing A. Necessarily $\tau(x) \supseteq \{A, \exists\mathsf{PP}.D_1, \exists\mathsf{PP}.D_2\}$.

Observation 3.2 (Strong arc-consistency). (V, D, τ) in Example 3.1 is strongly arc-consistent.

Proof. V has one element. There are no non-trivial pairs, so weak arc-consistency is vacuous.

For the existential clause, iterate over $\exists R.E \in \tau(x)$:

- $\exists\mathsf{PP}.D_1$: we need $\gamma_1 \in \mathrm{Tp}(C_0, T)$ with $D_1 \in \gamma_1$ and $\{F : \forall\mathsf{PP}.F \in \tau(x)\} \subseteq \gamma_1$. $\tau(x)$ contains no $\forall\mathsf{PP}$-universal (the TBox has no axioms of the form $A \sqsubseteq \forall\mathsf{PP}.\cdot$), so the forward-propagation set is empty. A Hintikka type γ_1 containing D_1 exists (take γ_1 saturating $\{D_1, \forall\mathsf{DR}.\neg D_2, \forall\mathsf{PO}.\neg D_2, \forall\mathsf{PP}.\neg D_2, \forall\mathsf{PPI}.\neg D_2, \neg D_2, \ldots\}$; this is internally consistent). The quantifier over $z \in V \setminus \{x\}$ is vacuous ($V \setminus \{x\} = \varnothing$), so the condition holds.
- $\exists\mathsf{PP}.D_2$: analogous. A Hintikka type γ_2 containing D_2 exists (e.g., $\{D_2, \neg D_1, \ldots\}$ saturating). Condition holds.

Both existentials pass. Strong arc-consistency holds. □

Observation 3.3 (Unrealisability). There is no model $M \models T$ containing an element x^M of type $\tau(x)$.

Proof. Suppose M were such a model. By (T1) and (T2) there exist $y_1, y_2 \in M$ with $\mathsf{PP}(x, y_i)$ and $D_i \in \mathrm{tp}^M(y_i)$ for $i = 1, 2$. By (T7) applied at y_1, $\neg D_2 \in \mathrm{tp}^M(y_1)$. Hence $y_1 \neq y_2$ (they differ in whether they contain D_2).

Consider $R := \rho^M(y_1, y_2) \in \{\mathsf{DR}, \mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}\}$.

- $R = \mathsf{DR}$: (T3) applied at y_1 (which has $\forall\mathsf{DR}.\neg D_2$) forces $\neg D_2 \in \mathrm{tp}^M(y_2)$; but $D_2 \in \mathrm{tp}^M(y_2)$. Contradiction.
- $R = \mathsf{PO}$: (T4) analogous.
- $R = \mathsf{PP}$: $\mathsf{PP}(y_1, y_2)$ means y_2 is y_1's PP-successor. (T5) at y_1 gives $\forall\mathsf{PP}.\neg D_2$, so y_2 has $\neg D_2$. Contradiction.
- $R = \mathsf{PPI}$: $\mathsf{PPI}(y_1, y_2)$ means y_1 is y_2's PP-successor, equivalently $\mathsf{PP}(y_2, y_1)$, equivalently y_2 has y_1 as a PP-successor. But we need (T6) on y_1: $\forall\mathsf{PPI}.\neg D_2$. PPI-successor of y_1 is y_2; so y_2 has $\neg D_2$. Contradiction.
- $R = \mathsf{EQ}$: $y_1 = y_2$, contradicting $y_1 \neq y_2$.

No relation is admissible. Hence no model. □

Corollary 3.4. *Proposition 5.2 of [2] is false: Example 3.1 is strongly arc-consistent but unrealisable.*

4 What went wrong

The gap is structurally similar to the one-step-up gap identified in [2], but one level deeper. Where the two-node counterexample exposed that *pending-witness vs. existing-node* conflicts are

not caught by weak arc-consistency, this one-node counterexample exposes that *pending-witness vs. pending-witness* conflicts are not caught by strong arc-consistency.

Strong arc-consistency checks each pending existential $\exists R.E \in \tau(x)$ by asking: does there exist a witness type γ such that γ connects to every existing node $z \in V$? It does not ask: does γ connect to every *other pending witness* γ' generated by some other pending existential?

In the one-node example, $V = \{x\}$ has no existing nodes other than x; so the existing-z quantifier is vacuous and every individually well-formed γ passes. The conflict is entirely between γ_1 (witness for $\exists \mathsf{PP}.D_1$) and γ_2 (witness for $\exists \mathsf{PP}.D_2$).

Remark 4.1 (Scaling pattern). The counterexample size has decreased with each correction.

- Weak $\to$ strong: the gap was between V and a single pending witness. The counterexample is 2 nodes, 7 axioms.
- Strong $\to$ stronger: the gap is between two pending witnesses at the same node. The counterexample is 1 node, 7 axioms.
- Stronger $\to$ even stronger would presumably be: gap between two pending witnesses at *different* nodes, requiring a compositionally-induced pair-domain. We expect this to have a slightly larger counterexample but the same structural flavour.

We do not attempt to construct the next-level example here; the pattern is clear enough to motivate the correct fix directly.

5 The fix: inter-witness pair-domains

The correct generalisation is to treat pending witnesses as virtual nodes and check pairwise type-safety among them. Concretely:

Definition 5.1 (Stronger arc-consistency)**.** A typed network (V, D, τ) is *stronger arc-consistent* if there exists an assignment

$$\gamma : \{(x, \exists R.E) : x \in V,\ \exists R.E \in \tau(x)\} \longrightarrow \mathrm{Tp}(C_0, T)$$

such that:

(i) Each $\gamma(x, \exists R.E)$ contains E and the forward-propagated universals $\{F : \forall R.F \in \tau(x)\}$.

(ii) For each $(x, \exists R.E)$ and each $z \in V \backslash \{x\}$: $\mathrm{comp}(\mathrm{inv}(R), D(x, z)) \cap \mathrm{TypeSafe}(\gamma(x, \exists R.E), \tau(z)) \neq \varnothing$.

(iii) For each pair of pending existentials $(x, \exists R_1.E_1) \neq (x', \exists R_2.E_2)$: the induced pair-domain $\mathrm{IPD} := \mathrm{IPD}(x, R_1, x', R_2) \cap \mathrm{TypeSafe}(\gamma(x, \exists R_1.E_1), \gamma(x', \exists R_2.E_2))$ is non-empty, where $\mathrm{IPD}(x, R_1, x', R_2)$ is the RCC5-composition-induced set of possible relations between the two pending witnesses (if $x = x'$: $\mathrm{IPD} = \mathrm{comp}(\mathrm{inv}(R_1), R_2)$; otherwise $\mathrm{IPD} = \mathrm{comp}(\mathrm{comp}(\mathrm{inv}(R_1), D(x, x')), R_2$

(iv) Weak arc-consistency holds on the original V.

Clause (iii) is the new condition. It says: once we commit to witness types for each pending existential, pairs of witnesses must be pairwise type-safe under the composition-induced domains.

Observation 5.2. Example 3.1 fails stronger arc-consistency.

Proof. For the two pending existentials $\exists\mathsf{PP}.D_1$ and $\exists\mathsf{PP}.D_2$ at x: $\mathrm{IPD}(x, \mathsf{PP}, x, \mathsf{PP}) = \mathrm{comp}(\mathsf{PPI}, \mathsf{PP}) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}\}$. For every choice $\gamma_1 \ni D_1$ and $\gamma_2 \ni D_2$: $\mathrm{TypeSafe}(\gamma_1, \gamma_2)$ filters as in Observation 3.3—every atom is forbidden by some $\forall R.\neg D_2$ universal in γ_1, and EQ is forbidden by type-disjointness. Hence $\mathrm{IPD} \cap \mathrm{TypeSafe} = \varnothing$ for every assignment, and clause (iii) fails. □

Proposition 5.3 (Conjectural; equivalent to split-forest rank-d validity)**.** *Stronger arc-consistency, augmented with propagation to pending witnesses of* pending witnesses *(recursively up to modal depth d of the concept), is equivalent to the split-forest paper's rank-d validity conditions on an implicit quotient [3].*

Sketch. Unfold the witness assignment γ recursively: witnesses of witnesses constitute rank-$(d-1)$ states relative to V's rank-d states. Clause (iii) for pending witnesses is exactly the pair-domain condition in the split-forest paper's Definition 1.10 of valid rank-d quotients. Conditions (i)–(ii) correspond to witness-menu relation-safety and $_R$-filtering. The correspondence is bookkeeping-exact modulo normalisation of sibling-interface descriptors (C1)–(C5), which are implied by clauses (iii) and their recursive extension. □

We do not fill in this sketch; the substance is in the split-forest paper's soundness proof [3]. The point of Proposition 5.3 is the observation that our three-step cascade terminates at split-forest validity—not at some strictly simpler propagation.

6 Implications

6.1 The cascade

Summarising the three-paper sequence:

Paper	Propagation	Gap caught	Counterexample size	Conjecture
[1]	weak AC	intra-V type-safety	—	Prop. 7.1 conj. (false)
[2]	strong AC	+ V-to-witness extensibility	2 nodes, 7 ax	Prop. 5.2 conj. (false)
this paper	stronger AC	+ witness-to-witness pair-domains	1 node, 7 ax	Prop. 5.3: = split-fores

The pattern:

- Each level checks one more layer of the witness-extension structure. Each simplification that omits a layer has a counterexample.
- Counterexample *size* decreases as the propagation strengthens, because each stronger definition catches more local failures, so the remaining gaps are those that exploit a purer version of the missing layer.
- The final propagation that catches everything is exactly split-forest rank-d validity: there is no shortcut.

6.2 For decidability

Decidability is unaffected. The split-forest soundness [3] and completeness extraction [4] together establish decidability of $\mathcal{ALCI}_{\mathsf{RCC5}}$ using the full rank-d validity conditions, not any of our simplified propagations. Our three counterexamples refute three proposed simplifications; they do not touch the underlying result.

6.3 For undecidability

The cascade is *further* evidence against undecidability, in the following sense. Every undecidability reduction attempt we have documented was blocked by some local consistency condition. The three papers together exhibit the full chain of consistency conditions that must be satisfied:

1. Pairwise type-safety on V (captured by weak AC).
2. Existential-witness extensibility from V (captured by strong AC).
3. Inter-witness pair-domain consistency (captured by stronger AC).
4. Recursive versions of (3) up to modal depth d (captured by split-forest rank-d validity).

Each level is a binary or pair-local constraint, generated mechanically from the TBox. None introduces a genuinely k-ary ($k \geq 4$) constraint in the sense of being irreducible to the lower levels. The cascade concretises the "no k-ary synthesis" intuition from the probe paper [1]: ALCI cannot force arity-4 or higher constraints because every constraint the TBox can express decomposes through the witness-extension hierarchy.

If there *is* a route to undecidability for $\mathcal{ALCI}_{\mathsf{RCC5}}$, it must sit outside this hierarchy. We do not see such a route.

6.4 For the split-forest paper

The apparent complexity of [3]'s rank-d machinery—witness menus, sibling-interface descriptors (C1)–(C5), $_R$-filtering, canonical refinements—is no longer discretionary. Each piece corresponds to a specific extension-consistency check that, if omitted, admits a counterexample of the shape documented in one of these three papers. The rank-d hierarchy is *the* minimal viable structure for completeness. Implementations (such as the cover-tree tableau [5]) that do not visibly perform all these checks are either empirically equivalent on the input distribution tested, or silently missing cases—a possibility that the methodological note in [2] flagged as the same distribution-blindness hazard that bit the pre-fix 911-test cross-validation.

7 Honest assessment

What this paper establishes:

1. A concrete one-node, seven-axiom counterexample (Example 3.1) that is strongly arc-consistent but unrealisable. The refutation of Proposition 5.2 of [2] is solid.
2. The correct fix (stronger arc-consistency, Definition 5.1) adds an inter-witness pair-domain clause and is conjecturally equivalent to split-forest rank-d validity.

3. The three-paper cascade exhibits a pattern: every simplification of split-forest validity has a counterexample, and the counterexamples shrink as propagation strengthens.

What we do not establish:

1. A proof of Proposition 5.3. The substance is in [3]; we have not extracted the equivalence rigorously.
2. An exhaustion of possible simplifications. There might be a different simpler propagation, not in the cascade we considered, that catches the three counterexamples simultaneously without going all the way to rank-d validity. We are sceptical but have not proved impossibility.
3. Any route to undecidability. The cascade is evidence against, not proof against.

Closing. Three attempts to find a simpler alternative to split-forest validity have failed, each with a strictly smaller counterexample than the last. We stop here. The next productive question for the project is probably not "is there an even simpler propagation" but "does strong empirical validation of the cover-tree tableau (which does *not* visibly perform inter-witness pair-domain checks) imply either (a) the cover-tree implicitly catches these cases through some other mechanism, or (b) the tested input distribution systematically excluded them?" That is a question about the cover-tree implementation, not about the split-forest semantics, and it is the right direction for the remaining theory-vs-implementation gap documented in the overview paper.

References

[1] Claude (Anthropic, Opus 4.7), prompted by M. Wessel. Patchwork preservation under $\mathcal{ALCI}_{\mathsf{RCC5}}$ augmentation: A probe toward undecidability. Manuscript, `papers/patchwork_augmentation_ALCIRCC5.pdf`, April 2026.

[2] Claude (Anthropic, Opus 4.7), prompted by M. Wessel. A two-node counterexample to naive typed patchwork. Manuscript, `papers/typed_patchwork_counterexample_ALCIRCC5.pdf`, April 2026.

[3] GPT-5.4 (OpenAI), prompted by M. Wessel. Sibling-interface descriptors and split-forest semantics for $\mathcal{ALCI}_{\mathsf{RCC5}}$. Manuscript, `papers/trees/sibling_interface_descriptors_ALCIRCC5_completed_eqsync_canonical_needpatch` April 2026.

[4] Claude (Anthropic, Opus 4.6), prompted by M. Wessel. Completeness extraction for $\mathcal{ALCI}_{\mathsf{RCC5}}$. Manuscript, `papers/completeness_extraction_ALCIRCC5.pdf`, April 2026.

[5] Claude (Anthropic, Opus 4.6 and Opus 4.7), prompted by M. Wessel. A cover-tree tableau for $\mathcal{ALCI}_{\mathsf{RCC5}}$. Manuscript, `papers/cover_tree_tableau_ALCIRCC5.pdf`, April 2026.